

The Curvature–Packing–Dynamics (CPD) Framework: A Geometric Foundation for Structure and Motion

Andrew Pasternak
Independent Researcher
Sydney, NSW, Australia
Email: turbulence1@live.com.au
ORCID: 0009-0006-0164-8601

November 25, 2025

Abstract

Curvature–Packing–Dynamics (CPD) is introduced as a geometric framework for understanding how complex systems form, maintain, and transform internal structure. CPD defines three primitive fields: curvature C , packing P , and flux Q , which interact to produce spatial and temporal organization in both physical and nonphysical environments. Unlike domain-specific models that rely on bespoke equations, CPD provides a conceptual geometry applicable to reaction–diffusion systems, turbulence, ecological dynamics, strategic games, growth processes, and financial markets. This article establishes the core variables, the minimal interaction rules, the geometric logic underlying emergence, and a catalogue of canonical examples showing the breadth of CPD’s descriptive power. The goal is to offer a unified theoretical foundation on which future CPD research can build formal models, simulations, and applications.

1 Introduction

Across scientific disciplines, complex systems display remarkably similar patterns: spatial domains emerge, boundaries form and dissolve, waves propagate, instabilities amplify, and structures appear where none existed. Examples include chemical spirals, ecological fronts, turbulent vortices, dendritic growth, coordinated flows in crowds, and strategic tension in games such as chess or Go.

Traditionally, each domain explains these phenomena using specialized mathematics—partial differential equations in physics, payoff matrices in game theory, reaction schemes

in chemistry, or optimization principles in economics. Yet the qualitative patterns share common features, suggesting that a deeper, geometry-based description may underlie them.

This article introduces the *Curvature–Packing–Dynamics* (CPD) framework as a minimal geometric language capable of expressing structural emergence across domains. CPD does not replace existing equations; instead, it reveals the geometric substrate common to them. The framework is built upon three interacting fields:

- curvature $C(x, t)$: local imbalance or potential shape,
- packing $P(x, t)$: density, congestion, or crowding,
- flux $Q(x, t)$: the flow or transport driven by C and modulated by P .

The central hypothesis is that the interaction of these three fields provides a universal description of structure, whether in physical, biological, informational, or strategic systems.

2 The CPD Variables

2.1 Curvature C

Curvature denotes the local shape or imbalance of an effective landscape governing how states change. Mathematically, curvature may manifest as the second derivative of a potential, the Laplacian of a field, or the directional asymmetry of transitions. Conceptually, curvature points to where the system “wants” to move or concentrate.

2.2 Packing P

Packing represents local density, occupancy, or exclusion. Regions of high P resist deformation or displacement, while regions of low P allow flux and curvature-driven movement. Packing is ubiquitous: molecular crowding, population density, liquidity concentration, or the spatial presence of chess pieces.

2.3 Flux Q

Flux captures the flow of material, probability, energy, information, or strategic pressure. It is the dynamical response to curvature gradients, with packing modulating the magnitude or direction of transport.

2.4 Minimal CPD triad

The foundational interaction cycle is:

Curvature C shapes where flux Q flows. **Flux** Q redistributes packing P .
Packing P modifies curvature C by concentrating or resisting structure.

This closed loop is hypothesized to be the generative mechanism behind a wide class of emergent patterns.

3 Conceptual Evolution Equations

CPD does not impose a single governing PDE. Instead, it provides a structural scaffold into which specific domain models can be embedded. The minimal functional forms are:

3.1 Curvature evolution

$$\frac{\partial C}{\partial t} = \mathcal{F}_C(P, Q, \nabla P, \nabla \cdot Q), \quad (1)$$

where the functional $\mathcal{F}_C$ captures local reactions, tension propagation, or feedback.

3.2 Packing evolution

$$\frac{\partial P}{\partial t} = \mathcal{F}_P(C, Q). \quad (2)$$

3.3 Flux representation

The simplest formulation defines flux as a gradient flow:

$$Q = -\nabla \Phi(C, P), \quad (3)$$

where Φ is an effective potential. More generally,

$$\frac{\partial Q}{\partial t} = \mathcal{F}_Q(C, P), \quad (4)$$

allowing inertial, stochastic, or strategic flux dynamics.

4 Mechanistic Logic of CPD

4.1 Curvature as directional incentive

High curvature reflects regions where the landscape is steep, encouraging motion or accumulation. In physical systems this may be a force; in strategic systems it may be tension or incentive.

4.2 Packing as local resistance

High packing slows or redirects flux. This geometric resistance underlies crowding, mechanical strain, or saturation.

4.3 Flux transports structure

Flux redistributes both curvature and packing, either smoothing imbalances or amplifying them depending on alignment.

4.4 Emergent structure from mismatch

Stable or metastable structures arise where curvature desires concentration but packing prevents further motion, such as wave fronts, boundaries, or vortices.

5 Canonical Examples

5.1 Reaction–diffusion

- C : chemical potential curvature
- P : concentration state
- Q : diffusive flux

Patterns appear where reaction-driven curvature and diffusion-driven flux reach a balance.

5.2 Turbulence

- C : curvature of kinetic–pressure field
- P : vorticity concentration or strain packing
- Q : velocity flux

Vortex sheets arise from curvature–packing mismatch.

5.3 Ecological fronts

- C : growth advantage or fitness curvature
- P : species density
- Q : migratory flux

5.4 Chess and strategic systems

- C : strategic pressure landscape
- P : piece congestion
- Q : move or threat propagation

CPD identifies critical lines and bottlenecks where curvature is high but packing blocks expansion.

5.5 Growth and morphogenesis

- C : chemical or mechanical potential
- P : material density
- Q : growth flux

6 Predictions and Structural Insights

CPD implies several general phenomena:

1. **Structure forms where curvature and packing conflict.**
2. **Pattern location can be predicted early** from C/P geometry.
3. **Instabilities arise** where flux aligns with curvature but encounters low packing.
4. **Boundaries appear** where flux is suppressed by high packing.

CPD also hints at unifying principles linking physical and informational systems through shared geometry.

7 Implications and Future Work

Future work includes:

- Developing CPD differential equations for specific domains.
- Constructing numerical solvers for CPD systems.
- Formally connecting CPD to optimal transport and geometric mechanics.
- Applying CPD to turbulence modeling, ecological networks, growth, and strategy.
- Incorporating CPD into machine learning architectures as a geometric prior.

CPD’s core value lies in offering a minimal, universal geometry of emergence grounded in three interacting fields.

References

1. Turing, A. (1952). “The Chemical Basis of Morphogenesis.” *Philosophical Transactions of the Royal Society B*.
2. Prigogine, I., Nicolis, G. (1977). *Self-Organization in Nonequilibrium Systems*.
3. Cross, M., Hohenberg, P. (1993). “Pattern formation outside equilibrium.” *Reviews of Modern Physics*.
4. Batchelor, G. (1967). *An Introduction to Fluid Dynamics*.
5. Pope, S. (2000). *Turbulent Flows*.
6. Murray, J. (2003). *Mathematical Biology*.
7. Nowak, M. (2006). *Evolutionary Dynamics*.
8. Meyer, G. (2020). “Geometry of Emergence and Spatial Competition.”
9. Villani, C. (2003). *Topics in Optimal Transportation*.
10. Shannon, C. (1948). “The Mathematical Theory of Communication.”
11. Silver, D. et al. (2016). “Mastering the game of Go with deep neural networks.” *Nature*.